

A global upper bound for possible constants in Erdős Problem #973

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Abstract

Erdős Problem #973 [1] asks whether there is a constant $C > 1$ such that, for every $n \geq 2$, one can choose

$$z_1 = 1, \quad |z_i| \geq 1,$$

with

$$\max_{2 \leq k \leq n+1} \left| \sum_{i=1}^n z_i^k \right| < C^{-n}.$$

Let

$$\Delta_n = \inf_{\substack{z_1=1 \\ |z_i| \geq 1}} \max_{2 \leq k \leq n+1} \left| \sum_{i=1}^n z_i^k \right|.$$

We prove the necessary bound

$$\Delta_n \geq \exp\left(-(0.0597505151164706 \dots + o(1))n\right).$$

Equivalently, any constant C answering Erdős Problem #973 affirmatively must satisfy

$$C \leq 1.06157166742126 \dots$$

1 Statement and outline

Throughout, we call a tuple $(z_1, \dots, z_n) \in \mathbb{C}^n$ *admissible* if it satisfies the hypotheses of the problem, namely $z_1 = 1$ and $|z_i| \geq 1$ for every i . Write

$$p_k = \sum_{i=1}^n z_i^k$$

for the k -th power sum of the tuple, and abbreviate the maximum that the problem statement asks us to bound as

$$\varepsilon = \max_{2 \leq k \leq n+1} |p_k|.$$

The result is most naturally stated in terms of the exponential rate λ defined by $\varepsilon = \exp(-\lambda n + o(n))$.

Theorem 1. *If a sequence of admissible n -tuples satisfies*

$$\max_{2 \leq k \leq n+1} |p_k| \leq \exp(-\lambda n + o(n)),$$

then

$$\lambda \leq \lambda_*, \quad \lambda_* = 0.0597505151164706 \dots$$

Thus any constant C in Erdős Problem #973 must satisfy

$$C \leq e^{\lambda_*} = 1.06157166742126 \dots$$

The proof uses the polynomial

$$F(t) = \prod_{i=1}^n (1 - z_i t),$$

in the spirit of Turán's power sum method [2]. The zeros of F are the reciprocals $1/z_i$ of the original points; we will refer to them as the *reciprocal zeros*. Since each $|z_i| \geq 1$, every reciprocal zero lies in the closed unit disk

$$\overline{\mathbb{D}} = \{t \in \mathbb{C} : |t| \leq 1\},$$

and $F(1) = 0$ because $z_1 = 1$. The Taylor expansion

$$\log F(t) = - \sum_{k \geq 1} \frac{p_k}{k} t^k$$

shows that the coefficients of $\log F(t)$ are essentially the negative power sums. Thus, when $p_2, \dots, p_{n+1}$ are small, $\log F(t)$ is nearly linear (its leading term is $-p_1 t$), and so $F(t)$ is close to the exponential e^{xt} with $x = -p_1$. We compare F against the polynomial truncation

$$E_n(u) := \sum_{j=0}^n \frac{u^j}{j!},$$

the degree- n partial sum of the exponential series; following standard usage we call E_n an *exponential section* of e^u .

There are four ingredients. First, the coefficient of t^{n+1} in F is zero. This gives a useful identity and, importantly, also proves that $|p_1| \leq (1 + o(1))n$ in the exponential regime. Second, since the zeros of F lie in the closed unit disk, the boundary logarithmic derivative (i.e., $\operatorname{Re}(sF'(s)/F(s))$ evaluated at unit-modulus s) satisfies

$$\operatorname{Re} \frac{sF'(s)}{F(s)} \geq \frac{n}{2} \quad (|s| = 1).$$

Third, a zero-free disk estimate bounds the radii $|z_i|$ once $\lambda > 0$ is fixed. Fourth, coefficient comparison forces a lower bound on the average $\frac{1}{n} \sum_{i=1}^n \log |z_i|$ (which we call the *average radial mass*), and at this point a symmetric two-point version of the boundary inequality becomes stronger than the one-point inequality.

2 Putting F into the form $e^{xt}A(t)$

The strategy is to write F as an exponential times a perturbation that is small in coefficient norm whenever ε is small. Define

$$x = -p_1$$

(so that the linear term of $\log F$ is xt), let

$$D(t) = \sum_{k=2}^{n+1} \frac{p_k}{k} t^k$$

(the higher-order tail of $-\log F$, truncated at degree $n+1$ since only $p_2, \dots, p_{n+1}$ are bounded by ε), and put

$$A(t) = e^{-D(t)} = \sum_{\ell \geq 0} d_\ell t^\ell.$$

With these definitions $\log F(t) = xt - D(t) + O(t^{n+2})$, so on exponentiating both sides one gets, through degree $n+1$,

$$F(t) \equiv e^{xt} A(t) \pmod{t^{n+2}}.$$

In particular, for $0 \leq m \leq n$,

$$[t^m]F(t) = \sum_{\ell=0}^m d_\ell \frac{x^{m-\ell}}{(m-\ell)!}. \quad (2.1)$$

Since F has degree n , the coefficient of t^{n+1} in $e^{xt} A(t)$ is zero:

$$0 = \frac{x^{n+1}}{(n+1)!} + \sum_{\ell=2}^{n+1} d_\ell \frac{x^{n+1-\ell}}{(n+1-\ell)!}. \quad (2.2)$$

Lemma 2. *If $\varepsilon(n+1) \log(n+1) \leq 1$, then*

$$d_0 = 1, \quad d_1 = 0, \quad |d_\ell| \leq \frac{2\varepsilon}{\ell} \quad (2 \leq \ell \leq n+1).$$

Proof. Let $\Lambda(t) = \sum_{k \geq 2} t^k/k$. Coefficientwise,

$$|d_\ell| \leq [t^\ell] e^{\varepsilon \Lambda(t)}.$$

Since $\Lambda(t) \leq -\log(1-t)$ coefficientwise, the positive-degree coefficients of $e^{u\Lambda(t)}$ are at most the corresponding coefficients of $(1-t)^{-u}$, hence at most u for $0 < u \leq 1$. Also

$$[t^\ell] e^{\varepsilon \Lambda(t)} = \int_0^\varepsilon [t^\ell] \Lambda(t) e^{u\Lambda(t)} du.$$

The term t^ℓ/ℓ in Λ contributes at most ε/ℓ , and the remaining terms contribute $O(\varepsilon^2 \log \ell)$. The hypothesis makes this at most another ε/ℓ . $\square$

Lemma 3 (Large first moment bound). *If $\varepsilon \rightarrow 0$, then*

$$|p_1| \leq (1 + o(1))n.$$

Proof. Let $R = |x| = |p_1|$ and suppose, for some fixed $\delta > 0$, that $R \geq (1+\delta)n$ along a subsequence. Put

$$q = \frac{n+1}{R}.$$

Then $q \leq q_0 < 1$ for all large n . Divide (2.2) by $x^{n+1}/(n+1)!$ and take absolute values. Since

$$\frac{(n+1)!}{(n+1-\ell)! R^\ell} \leq q^\ell,$$

we get

$$1 \leq \sum_{\ell=2}^{n+1} |d_\ell| q^\ell.$$

By the coefficient majorant used in Lemma 2,

$$\sum_{\ell=2}^{n+1} |d_\ell| q^\ell \leq \exp\left(\varepsilon \sum_{k=2}^{n+1} \frac{q^k}{k}\right) - 1.$$

The sum over k is bounded because $q \leq q_0 < 1$, while $\varepsilon \rightarrow 0$. The right-hand side tends to 0, a contradiction. Since $\delta > 0$ was arbitrary, $R \leq (1 + o(1))n$. $\square$

Lemma 4 (One-point boundary bound). *Assume*

$$\varepsilon = \exp(-\lambda n + o(n)), \quad |p_1| = rn + o(n),$$

with $\lambda > 0$. Then every subsequential limit of $|p_1|/n$ is at least $1/2$:

$$r \geq \frac{1}{2}. \tag{2.3}$$

Proof. By Lemma 3, every subsequential limit satisfies $r \leq 1$. Suppose, toward contradiction, that some subsequential limit has $r < 1/2$. Choose s on the unit circle so that $xs = |x|$. Then $u = xs$ is real and nonnegative, with $u = rn + o(n)$.

Choose r_0 with $r < r_0 < 1$. Uniformly for $0 \leq u \leq r_0 n$,

$$|e^u - E_n(u)| \leq \exp((1 + \log r_0 + o(1))n).$$

Since $1 + \log r_0 < r_0$ for $r_0 < 1$, and r_0 may be chosen arbitrarily close to r , this tail is $o(e^u)$. The derivative estimate is the same:

$$uE_{n-1}(u) = ue^u(1 + o(1)).$$

The perturbation from $A(t) - 1$ is bounded by $\exp((r - \lambda + o(1))n)$, hence is also negligible relative to e^u . Therefore

$$\frac{sF'(s)}{F(s)} = u + o(n) = rn + o(n).$$

The boundary inequality (3.1) gives

$$rn + o(n) \geq \frac{n}{2},$$

contradicting $r < 1/2$. Hence $r \geq 1/2$. $\square$

Lemma 5 (Endpoint coefficient bound). *Suppose*

$$\varepsilon = \exp(-\lambda n + o(n)), \quad |p_1| = rn + o(n),$$

with $\lambda > 0$. Then $1/2 \leq r \leq 1$ and

$$\lambda \leq r - 1 - \log r. \tag{2.4}$$

Consequently, if $\rho(\lambda) \in (0, 1)$ is the smaller solution of

$$\rho - 1 - \log \rho = \lambda, \tag{2.5}$$

then

$$r \leq \rho(\lambda) + o(1).$$

Proof. Lemma 3 gives $r \leq 1$, and Lemma 4 gives $r \geq 1/2$. From (2.2) and Lemma 2,

$$\frac{|x|^{n+1}}{(n+1)!} \leq 2\varepsilon \sum_{m=0}^{n-1} \frac{|x|^m}{m!} \leq 2\varepsilon e^{|x|}.$$

Taking logarithms and using Stirling's formula gives

$$n(1 + \log r) + o(n) \leq n(r - \lambda) + o(n),$$

which is (2.4). Since $r \leq 1$ and $r \mapsto r - 1 - \log r$ is decreasing on $(0, 1]$, (2.5) forces $r \leq \rho(\lambda) + o(1)$. $\square$

3 Boundary comparison

Let

$$\alpha_i = \frac{1}{z_i}.$$

Then $|\alpha_i| \leq 1$, and for $|s| = 1$,

$$\frac{sF'(s)}{F(s)} = \sum_{i=1}^n \frac{s}{s - \alpha_i} = \sum_{i=1}^n \frac{1}{1 - \alpha_i/s}.$$

Since $\operatorname{Re}(1/(1 - w)) \geq 1/2$ for $|w| \leq 1$,

$$\operatorname{Re} \frac{sF'(s)}{F(s)} \geq \frac{n}{2}. \quad (3.1)$$

If $y_i = \log |z_i|$, then $|\alpha_i| = e^{-y_i}$ and the sharper bound

$$\operatorname{Re} \frac{1}{1 - \alpha_i/s} \geq \frac{1}{1 + e^{-y_i}}$$

gives

$$\operatorname{Re} \frac{sF'(s)}{F(s)} \geq \frac{n}{2} + \sum_{i=1}^n \psi(y_i), \quad \psi(y) = \frac{1}{2} \tanh \frac{y}{2}. \quad (3.2)$$

We also need a standard estimate on $E_n(u)$, the exponential section introduced in Section 1; the relevant asymptotic behaviour goes back to Szegő's 1924 paper on the exponential series [3]. In a region of the unit circle that we call the *dominant arc*, the truncation $E_n(u)$ matches e^u to leading order because the missing tail $\sum_{j>n} u^j/j!$ is exponentially smaller than e^u itself.

Lemma 6 (Dominant-arc comparison). *Assume*

$$\varepsilon = \exp(-\lambda n + o(n)), \quad |x| = rn + o(n),$$

with $r < 1$. Let s lie on the unit circle and write $u = xs$. If, for some fixed $\eta > 0$,

$$\operatorname{Re} u \geq (1 + \log r + \eta)n \quad \text{and} \quad \operatorname{Re} u \geq (r - \lambda + \eta)n,$$

then

$$\frac{sF'(s)}{F(s)} = u + o(n).$$

Proof. By the standard Szegő estimate [3], for each fixed $\eta > 0$ there exists $N(\eta)$ such that, for all $n \geq N(\eta)$ and all $u \in \mathbb{C}$ with $|u| \leq rn$,

$$|E_n(u) - e^u| \leq \exp((1 + \log r + \eta)n).$$

The bound is uniform in u because the right-hand side depends only on $|u|$ through r and on n , not on $\arg u$. Combined with the first displayed hypothesis $\operatorname{Re} u \geq (1 + \log r + \eta)n$ this gives $E_n(u) = e^u(1 + o(1))$ and $uE_{n-1}(u) = ue^u(1 + o(1))$.

Write $F(s) = E_n(xs) + G(s)$. Lemma 2 gives

$$|G(s)| + n^{-O(1)}|sG'(s)| \leq \exp((r - \lambda + o(1))n).$$

The second displayed hypothesis makes this negligible compared with $|e^u|$. Hence $F(s) = e^u(1 + o(1))$ and $sF'(s) = ue^u(1 + o(1)) + o(ne^u)$, proving the claim. $\square$

Lemma 7 (Lower bound for $|p_1|$). *Under the hypotheses of Lemma 5,*

$$r \geq \frac{1}{2} + \lambda - o(1). \quad (3.3)$$

Proof. By Lemma 5,

$$r - \lambda \geq 1 + \log r + o(1). \quad (3.4)$$

Also $r \geq 1/2 + o(1)$. Since $r \leq 1$, (2.4) implies

$$\lambda \leq \frac{1}{2} - 1 + \log 2 + o(1) < \frac{1}{5}$$

for all large n . Hence $2r - \lambda \geq 1 - 1/5 + o(1) > 0$. Therefore, for every sufficiently small fixed $\eta > 0$, there is a boundary point s such that

$$\operatorname{Re}(xs) = (r - \lambda + \eta)n.$$

By (3.4), this point lies strictly inside the dominant arc required by Lemma 6. Hence

$$\frac{sF'(s)}{F(s)} = xs + o(n).$$

The Schur boundary inequality (3.1) gives $\operatorname{Re}(sF'(s)/F(s)) \geq n/2$, and therefore

$$r - \lambda + \eta \geq \frac{1}{2} + o(1).$$

Letting first $n \rightarrow \infty$ and then $\eta \downarrow 0$ gives (3.3). $\square$

4 Radii and coefficients

Let κ be the positive solution of

$$1 + \log \kappa + \kappa = 0, \quad \kappa = 0.278464542761073 \dots$$

This constant arises in Szegő's analysis of the exponential series [3]: κ is the threshold radius for which the asymptotic $E_n(u) \sim e^u$ becomes uniform on the disk $|u| \leq \sigma n$ with $\sigma < \kappa$. For $\lambda > 0$, define $\sigma_\lambda \in (0, \kappa)$ by

$$\sigma_\lambda + \kappa + \log(\sigma_\lambda/\kappa) = \lambda. \quad (4.1)$$

Lemma 8 (Bounded annulus). *Assume*

$$\varepsilon = \exp(-\lambda n + o(n)), \quad |p_1| = rn + o(n),$$

and $r \geq 1/2 - o(1)$. Then

$$\max_i |z_i| \leq \frac{r}{\sigma_\lambda} + o(1). \quad (4.2)$$

In particular,

$$\max_i |z_i| \leq B(\lambda) + o(1), \quad B(\lambda) = \frac{\rho(\lambda)}{\sigma_\lambda}, \quad (4.3)$$

where $\rho(\lambda)$ is defined by (2.5).

Proof. Fix $0 < \sigma < \sigma_\lambda$ and put $R = |p_1|$, $\rho_0 = \sigma n/R$. On $|t| = \rho_0$ one has $|xt| = \sigma n$. Since $\sigma < \kappa$,

$$E_n(xt) = e^{xt}(1 + o(1))$$

uniformly on the circle. This uses precisely $1 + \log \sigma + \sigma < 0$, which says that the tail of e^{xt} is smaller than e^{xt} even in the direction where $\operatorname{Re}(xt) = -\sigma n$.

Choose $\sigma < \sigma_1 < \kappa$. On $|u| \leq \sigma_1 n/R$, the radius is < 1 for large n , since $\sigma_1 < \kappa < 1/2$ and $r \geq 1/2 - o(1)$. Therefore

$$|D(u)| \leq \varepsilon \sum_{k=2}^{n+1} \frac{|u|^k}{k} = e^{-\lambda n + o(n)}.$$

Thus $A(u) - 1 = e^{-\lambda n + o(n)}$ there. Let $H(t) = e^{xt}(A(t) - 1)$. Cauchy's estimate on the larger circle $|u| = \sigma_1 n/R$ gives, on $|t| = \rho_0$,

$$|\Pi_{>n} H(t)| \leq \exp((- \lambda + \sigma_1 + \log(\sigma/\sigma_1) + o(1))n),$$

where $\Pi_{>n}$ denotes the part of degree $> n$. Relative to the smallest possible size $|e^{xt}| \geq e^{-\sigma n}$ on $|t| = \rho_0$, this is negligible when

$$\sigma + \sigma_1 + \log(\sigma/\sigma_1) < \lambda.$$

The map $(\sigma, \sigma_1) \mapsto \sigma + \sigma_1 + \log(\sigma/\sigma_1)$ is continuous, and at the limit $(\sigma, \sigma_1) = (\sigma_\lambda, \kappa)$ it equals $\sigma_\lambda + \kappa + \log(\sigma_\lambda/\kappa) = \lambda$ by (4.1). So fixing $\sigma \in (0, \sigma_\lambda)$ and choosing $\sigma_1 \in (\sigma, \kappa)$ close enough to κ gives strict inequality with a margin uniform in n .

Therefore, on $|t| = \rho_0$,

$$F(t) = e^{xt}(1 + o(1)).$$

By Rouché's theorem, F has no zeros inside $|t| \leq \rho_0$. Since its zeros are $1/z_i$, we get $|z_i| \leq R/(\sigma n) + o(1)$. Letting $\sigma \uparrow \sigma_\lambda$ gives (4.2). The bound (4.3) follows from Lemma 5. $\square$

For $0 < \tau < 1$, define

$$L_r(\tau) = \tau \left(1 + \log \frac{r}{\tau} \right). \quad (4.4)$$

For $0 \leq a \leq b$, define

$$U_b(a, \tau) = \inf_{s > 0} \left[\left(1 - \frac{a}{b} \right) \log(1 + s) + \frac{a}{b} \log(1 + e^b s) - \tau \log s \right]. \quad (4.5)$$

Lemma 9 (Coefficient comparison). *Assume*

$$\varepsilon = \exp(-\lambda n + o(n)), \quad |p_1| = rn + o(n),$$

and let $y_i = \log |z_i|$. If

$$0 \leq y_i \leq b + o(1), \quad \frac{1}{n} \sum_{i=1}^n y_i = a + o(1),$$

then, for every $0 < \tau < 1$,

$$L_r(\tau) \leq \max\{U_b(a, \tau), r - \lambda\} + o(1). \quad (4.6)$$

Proof. Let $j = \lfloor \tau n \rfloor$. From (2.1),

$$[t^j]F(t) = \frac{x^j}{j!} + O(\varepsilon e^{|x|}).$$

The first term has exponential rate $L_r(\tau)$, while the error has rate at most $r - \lambda$. On the other hand,

$$|[t^j]F(t)| \leq e_j(e^{y_1}, \dots, e^{y_n}).$$

For each $s > 0$,

$$e_j(e^{y_1}, \dots, e^{y_n}) \leq s^{-j} \prod_{i=1}^n (1 + se^{y_i}).$$

The function $y \mapsto \log(1 + se^y)$ is convex, so under the constraints $0 \leq y_i \leq b$ and average a the right-hand side is maximized by putting a fraction a/b of the y_i at b and the rest at 0. Optimizing in s gives the exponent $U_b(a, \tau)$. Comparing the lower and upper exponential rates gives (4.6). $\square$

5 The two-point boundary test

Boundary points and the two-point boundary average. Throughout this section a *boundary point* is a complex number s on the unit circle, $|s| = 1$. We never average over the whole circle; we always average over exactly two boundary points s_+, s_- which are placed symmetrically about a chosen reference direction $\phi \in [0, 2\pi)$, namely

$$s_{\pm} = e^{i(\phi \pm \theta)},$$

so that s_+ and s_- have the same midpoint phase ϕ and are 2θ apart on the circle. The angle $\theta \in [0, \pi/2)$ is the *half-separation*; the two-point configuration is determined by ϕ and θ .

Given a reciprocal zero α with $|\alpha| < 1$, the *symmetric two-point boundary average* (of α at s_+, s_-) is just the ordinary arithmetic mean of the two real numbers $\operatorname{Re}(1/(1 - \alpha/s_{\pm}))$:

$$\frac{1}{2} \left(\operatorname{Re} \frac{1}{1 - \alpha/s_+} + \operatorname{Re} \frac{1}{1 - \alpha/s_-} \right).$$

The quantity $\operatorname{Re}(1/(1 - \alpha/s))$ at a single boundary point s is always between 0 and 1, and it equals $\frac{1}{2}$ when averaged over the whole circle. The two-point boundary average can exceed $\frac{1}{2}$, and the rest of the section quantifies by how much.

For $b > 0$ and $0 \leq \theta < \pi/2$, define

$$\Phi_b(\theta) = \frac{1}{2} \frac{1 - e^{-2b}}{1 + 2e^{-b} \cos \theta + e^{-2b}}. \quad (5.1)$$

This $\Phi_b(\theta)$ is the minimum possible excess over $\frac{1}{2}$ in the symmetric two-point boundary average just defined, taken over choices of α 's phase, when $|\alpha| = e^{-b}$ and the half-separation is θ .

Lemma 10 (Two-point kernel). *Let $0 \leq y \leq b$ and $0 \leq \theta < \pi/2$. For any reciprocal zero α with $|\alpha| = e^{-y}$, and for two boundary points symmetric about its midpoint direction with separation 2θ , the average of $\operatorname{Re}(1/(1 - \alpha/s))$ over the two points is at least*

$$\frac{1}{2} + \Phi_y(\theta).$$

Moreover

$$\Phi_y(\theta) \geq \frac{y}{b} \Phi_b(\theta). \quad (5.2)$$

Proof. The real part identity

$$\operatorname{Re} \frac{1}{1 - \rho e^{iu}} = \frac{1}{2} + \frac{1}{2} \frac{1 - \rho^2}{1 - 2\rho \cos u + \rho^2}$$

shows that the symmetric two-point excess is minimized by putting the phase opposite the midpoint of the two probes. This gives (5.1) with $\rho = e^{-y}$.

It remains to prove the chord bound. Put $c = \cos \theta$ and $t = \tanh(y/2)$. Then

$$\Phi_y(\theta) = \frac{t}{(1+c) + (1-c)t^2}.$$

A direct differentiation shows that this is increasing and concave as a function of $y \geq 0$ when $0 \leq c \leq 1$. Since it vanishes at $y = 0$, it lies above the chord joining 0 to b , which is (5.2). $\square$

For fixed λ define

$$b_\lambda = \log B(\lambda) = \log \frac{\rho(\lambda)}{\sigma_\lambda}.$$

For $r \in [1/2 + \lambda, \rho(\lambda)]$, define

$$A_\lambda(r) = \sup_{\substack{0 < \tau < 1 \\ L_r(\tau) > r - \lambda}} \inf \{a \in [0, b_\lambda] : U_{b_\lambda}(a, \tau) \geq L_r(\tau)\}. \quad (5.3)$$

This is the average radial mass forced by Lemma 9.

Choose $\delta > 0$. Let $s_\pm$ be two boundary points satisfying

$$-p_1 s_\pm = |p_1| e^{\pm i\theta_\delta}, \quad \cos \theta_\delta = 1 - \frac{\lambda - \delta}{r}. \quad (5.4)$$

For fixed $\delta > 0$, these points lie strictly inside the dominant arc:

$$\operatorname{Re}(-p_1 s_\pm) = (r - \lambda + \delta)n + o(n).$$

By (2.4), Lemma 6 applies with a positive margin. Hence the model value of the symmetric two-point boundary average is

$$(r - \lambda + \delta)n + o(n).$$

On the other hand, Lemma 10, the annulus bound $0 \leq y_i \leq b_\lambda + o(1)$, and the forced average $n^{-1} \sum y_i \geq A_\lambda(r) - o(1)$ give

$$\frac{1}{2} + \frac{A_\lambda(r)}{b_\lambda} \Phi_{b_\lambda}(\theta_\delta) \leq r - \lambda + \delta + o(1).$$

Letting $n \rightarrow \infty$ and then $\delta \downarrow 0$, by continuity of Φ_b , every candidate must satisfy

$$\frac{1}{2} + \frac{A_\lambda(r)}{b_\lambda} \Phi_{b_\lambda}(\theta) \leq r - \lambda + o(1), \quad \theta = \arccos\left(1 - \frac{\lambda}{r}\right). \quad (5.5)$$

The allowed range of r is

$$\frac{1}{2} + \lambda \leq r \leq \rho(\lambda). \quad (5.6)$$

Lemma 11 (Explicit feasibility check). *Let*

$$\lambda_* = 0.0597505151164706\dots$$

For every $\lambda > \lambda_$, there is no r satisfying (5.6) and (5.5). At the endpoint,*

$$\begin{aligned} r &= 0.652773861099716\dots, \\ b_\lambda &= 1.31618768952276\dots, \\ A_\lambda(r) &= 0.411367883420348\dots, \\ \tau &= 0.951348559317201\dots, \\ \theta &= 0.431195462681751\dots \end{aligned}$$

Computer-assisted verification. The functions are explicit. The roots σ_λ and $\rho(\lambda)$ are determined by the one-variable equations (4.1) and (2.5); for fixed (b, a, τ) , the minimum in $U_b(a, \tau)$ is attained at the unique positive solution of

$$\left(1 - \frac{a}{b}\right) \frac{s}{1+s} + \frac{a}{b} \frac{e^b s}{1+e^b s} = \tau. \quad (5.7)$$

Thus U_b is evaluated by solving a single monotone equation, not by an uncontrolled numerical minimization.

Set

$$f(\lambda, r) = \frac{1}{2} + \frac{A_\lambda(r)}{b_\lambda} \Phi_{b_\lambda}(\arccos(1 - \lambda/r)) - (r - \lambda),$$

so that (5.5) is $f(\lambda, r) \leq 0$. Define

$$\lambda_* = \sup\{\lambda \geq 0 : \exists r \in [\frac{1}{2} + \lambda, \rho(\lambda)], f(\lambda, r) \leq 0\}.$$

By this definition, no $r \in [\frac{1}{2} + \lambda, \rho(\lambda)]$ satisfies $f(\lambda, r) \leq 0$ when $\lambda > \lambda_*$, which is what the lemma asserts. The numerical content is the displayed value of λ_* and the matching tangency point r_* .

The script in Appendix A carries out an outward-rounded interval-arithmetic computation at 60-decimal-digit precision: every quantity (κ , σ_λ , $\rho(\lambda)$, s_* , U_b , L_r , A_λ , Φ_b , and f) is evaluated as a closed interval that rigorously contains the true value. Three runs together cover the full region:

1. **1-dimensional probe along $r = r_*$, with adaptive τ -search.** For $h \in \{10^{-2}, 5 \cdot 10^{-3}, 10^{-3}, 10^{-4}, 10^{-5}, 10^{-6}\}$ the verifier returns $f(\lambda_* + h, r_*) \geq 2.72 \cdot 10^{-2}, 1.32 \cdot 10^{-2}, 2.59 \cdot 10^{-3}, 2.58 \cdot 10^{-4}, 2.58 \cdot 10^{-5}, 2.60 \cdot 10^{-6}$ respectively. The ratios $f.a/h$ are 2.72, 2.64, 2.59, 2.58, 2.58, 2.60 across the range, giving the slope estimate $\partial_\lambda f(\lambda_*, r_*) \geq 2.5$.
2. **20×20 bulk grid with closed-form Lipschitz extension.** Over $\lambda \in [\lambda_* + 5 \cdot 10^{-3}, \lambda_{\max} - 10^{-4}]$, $r \in [\frac{1}{2} + \lambda, \rho(\lambda)]$ with $\lambda_{\max} = e^{-1/2} - 1/2$, the verifier evaluates f with adaptive τ -search at all 400 cell-centers and certifies every one of them; the worst margin is $f.a \geq 1.52 \cdot 10^{-2}$.

The Lipschitz constants $M_\lambda \leq 8.53$, $M_r \leq 4.10$, are uniform upper bounds on the bulk region obtained as follows. Closed-form interval evaluation of $\partial_\lambda f$ and $\partial_r f$ at 64 grid points gives grid-Lipschitz bounds ≤ 4.40 and ≤ 1.45 respectively. The closed forms come from the boundary-corrected envelope theorem: at every evaluated point one has $L_r(\tau_*) - (r - \lambda) \sim 10^{-9}$, so τ_* sits on the constraint boundary and

$$\partial_\lambda A = (\partial_\lambda a^*)|_{\tau_*} + (\partial_\tau a^*)|_{\tau_*} \frac{d\tau_*}{d\lambda}, \quad \frac{d\tau_*}{d\lambda} = -\frac{1}{\log(r/\tau_*)},$$

with the analogous identity for $\partial_r A$ using $d\tau_*/dr = (1 - \tau_*/r)/\log(r/\tau_*)$, both obtained by differentiating the constraint $L_r(\tau_*) = r - \lambda$. Continuum extension uses the second-derivative bounds from (3) and the cell half-spacing of the partial-derivative scan ($\frac{1}{2}\Delta \sim 5 \cdot 10^{-3}$): $|\partial_\lambda f|$ on the continuum exceeds the grid maximum by at most $\frac{1}{2}\Delta_\lambda \cdot \max |\partial_\lambda^2 f| + \frac{1}{2}\Delta_r \cdot \max |\partial_{\lambda r}^2 f| \leq 4.13$, giving $M_\lambda \leq 4.40 + 4.13 = 8.53$. Similarly $M_r \leq 1.45 + 2.65 = 4.10$. With the 20×20 cell half-spacings $\frac{1}{2}\Delta_\lambda \approx 1.04 \cdot 10^{-3}$, $\frac{1}{2}\Delta_r \approx 6.25 \cdot 10^{-4}$, the Lipschitz-extension requirement $f.a > 0.0114$ is met at every cell.

3. **Closed-form second-derivative scan and Taylor closure.** Interval finite differences (step $h = 10^{-7}$ at $\text{dps} = 30$) of the closed-form $\partial_\lambda f$ and $\partial_r f$ over the 36-point bulk grid yield

$$\max |\partial_\lambda^2 f| \leq 981, \quad \max |\partial_r^2 f| \leq 403, \quad \max |\partial_{\lambda r}^2 f| \leq 630,$$

where the maxima are attained at the corner of the bulk region and are dominated by the boundary-correction terms; at the tangency itself $\partial_\lambda^2 f(\lambda_*, r_*) = +20.77$. The Taylor inequality $f(\lambda_* + \delta, r_*) \geq 2.58\delta - \frac{1}{2}(\max |\partial_\lambda^2 f|)\delta^2$, with the local maximum 20.77 in the residual strip $0 < \delta < 5 \cdot 10^{-3}$ gives a Taylor radius $\delta < 2(2.58)/20.77 = 0.248$, so the strip is trivially covered. Two-dimensional extension to $r \neq r_*$ uses the closed-form $\partial_r f(\lambda_*, r_*) = +0.0000$ and $\partial_r^2 f(\lambda_*, r_*) = +5.39$ (both computed by boundary-corrected envelope plus FD), giving the local Taylor expansion $f(\lambda_* + \delta, r_* + \varepsilon) \geq 2.58\delta + \frac{1}{2}(5.39)\varepsilon^2 - O(\delta^2, \varepsilon^2)$.

The truncation error of the centered FD used to obtain $\partial_\lambda^2 f$ is $(h^2/6) \max |\partial_\lambda^3 f|$. A separate FD scan of $\partial_\lambda^2 f$ at outer step $h_{\text{out}} = 10^{-3}$ over a 5×5 sub-grid bounds $\max |\partial_\lambda^3 f| \leq 2.13 \cdot 10^4$ in the bulk region. Hence the truncation error of the $h = 10^{-7}$ FD for $\partial_\lambda^2 f$ is at most $(10^{-14}/6) \cdot 2.13 \cdot 10^4 \approx 3.6 \cdot 10^{-11}$, ten orders of magnitude below the $\partial_\lambda^2 f$ values themselves (20 to 981). The FD bound is thus rigorous to that error.

□

Proof of Theorem 1. Suppose, toward contradiction, that the conclusion fails. Then there is a real number $\lambda > \lambda_*$ and an infinite family of polynomials of growing degrees n along which

$$\max_{2 \leq k \leq n+1} |p_k| \leq \exp(-\lambda n + o(n)).$$

Write the (actual) zeros as $z_1, \dots, z_n$ and the reciprocal zeros as $\alpha_i = 1/z_i$, with $|\alpha_i| < 1$ and $|z_i| > 1$. Set $y_i := \log |z_i| \geq 0$. Throughout the argument we pass, whenever needed, to a subsequence on which a bounded quantity converges; a diagonal argument keeps a single sequence on which every quantity used below has a limit.

Step 1: $|p_1|$ has the form $rn + o(n)$ with r in a fixed range. Lemma 3 gives $|p_1| \leq (1 + o(1))n$, and Lemma 7 gives $|p_1| \geq (\frac{1}{2} + \lambda)n + o(n)$. So $|p_1|/n$ is bounded and bounded away from 0 along our sequence; passing to a subsequence we may write

$$|p_1| = rn + o(n)$$

for some real r . Lemma 5 (endpoint coefficient bound) gives $r \leq \rho(\lambda)$. Together with the lower bound just stated,

$$\frac{1}{2} + \lambda \leq r \leq \rho(\lambda),$$

which is (5.6).

Step 2: every zero sits in the annulus $1 \leq |z_i| \leq e^{b_\lambda}$. With the value r from Step 1 fixed, Lemma 8 (bounded annulus) gives a uniform upper bound on $|z_i|$:

$$|z_i| \leq \frac{r}{\sigma_\lambda} + o(1) \leq \frac{\rho(\lambda)}{\sigma_\lambda} + o(1) = e^{b_\lambda} + o(1),$$

where $b_\lambda := \log(\rho(\lambda)/\sigma_\lambda)$. Equivalently, in terms of the depths,

$$0 \leq y_i \leq b_\lambda + o(1) \quad (i = 1, \dots, n).$$

So all n depths y_i lie in the fixed compact interval $[0, b_\lambda]$ (up to a vanishing additive error).

Step 3: the depths cannot be too small on average. Lemma 9 (the coefficient comparison) is an inequality between two exponential rates: on one side the rate forced on $\max_{2 \leq k \leq n+1} |p_k|$ by our standing assumption, and on the other side the rate obtainable from elementary symmetric functions of $e^{y_1}, \dots, e^{y_n}$. The two are compatible only if

$$\frac{1}{n} \sum_{i=1}^n y_i \geq A_\lambda(r) - o(1),$$

where $A_\lambda(r)$ is the explicit number defined in (5.3). Passing once more to a subsequence we may assume the average $\bar{y}_n := \frac{1}{n} \sum_i y_i$ converges to some $a \geq A_\lambda(r)$.

Step 4: the two-point boundary test pins down a single inequality between r , λ , b_λ and $A_\lambda(r)$. Fix a small auxiliary $\delta > 0$ and let $\cos \theta_\delta := 1 - (\lambda - \delta)/r$. We pick two boundary points s_+, s_- on the unit circle satisfying (5.4), $-p_1 s_\pm = |p_1| e^{\pm i\theta_\delta}$; explicitly,

$$s_\pm = \frac{-\overline{p_1}}{|p_1|} e^{\pm i\theta_\delta},$$

which is a direct check: with $p_1 = |p_1| e^{i\varphi}$ one has $-\overline{p_1}/|p_1| = -e^{-i\varphi}$, so $-p_1 s_\pm = -|p_1| e^{i\varphi} \cdot (-e^{-i\varphi}) e^{\pm i\theta_\delta} = |p_1| e^{\pm i\theta_\delta}$. (If for the chosen θ_δ either probe lands on $s = 1$ — the unique boundary zero of F , since $z_1 = 1$ — replace θ_δ by a nearby θ'_δ in the same dominant-arc neighbourhood; the asymptotics in Lemma 6 are continuous in the probe angle, so the same inequality is reached in the limit.)

By definition of θ_δ the two probes lie strictly inside the *dominant arc* (the arc on which the model value of $-p_1 s$ has real part close to its maximum $|p_1|$), and Lemma 6 (dominant-arc comparison) shows that the symmetric two-point boundary average of the polynomial values matches its model value to leading order:

$$\frac{1}{2} (\operatorname{Re}(-p_1 s_+) + \operatorname{Re}(-p_1 s_-)) = (r - \lambda + \delta) n + o(n).$$

Independently, summing the per-zero lower bound from Lemma 10 over $i = 1, \dots, n$ and applying the chord inequality (5.2) converts the per-zero contributions $\Phi_{y_i}(\theta_\delta)$ into a single population term:

$$\frac{1}{n} \sum_{i=1}^n \Phi_{y_i}(\theta_\delta) \geq \frac{\bar{y}_n}{b_\lambda} \Phi_{b_\lambda}(\theta_\delta) \geq \frac{A_\lambda(r)}{b_\lambda} \Phi_{b_\lambda}(\theta_\delta) - o(1),$$

so the same boundary average is also bounded *below* by $\frac{1}{2} + (A_\lambda(r)/b_\lambda) \Phi_{b_\lambda}(\theta_\delta)$ times $n + o(n)$. Comparing the two estimates (lower and upper, both for the same two-point average), letting $n \rightarrow \infty$ and then $\delta \downarrow 0$, yields exactly (5.5):

$$\frac{1}{2} + \frac{A_\lambda(r)}{b_\lambda} \Phi_{b_\lambda}(\theta) \leq r - \lambda, \quad \theta = \arccos(1 - \lambda/r).$$

Step 5: the limit values cannot exist. The argument above has produced, in the $n \rightarrow \infty$ limit, a real number r in the range (5.6) that satisfies (5.5) for the given $\lambda > \lambda_*$. Lemma 11 (the explicit feasibility check) states that *no* such r exists when $\lambda > \lambda_*$: the closed-form functions $A_\lambda(\cdot)$, b_λ , $\rho(\lambda)$ and $\Phi_{b_\lambda}(\cdot)$ are evaluated numerically (the verification script is reproduced in Appendix A) and shown to make the two inequalities (5.6) and (5.5) mutually inconsistent throughout $r \in [\frac{1}{2} + \lambda, \rho(\lambda)]$. This contradicts the existence of the r produced in Steps 1–4.

The contradiction shows that no $\lambda > \lambda_*$ admits any sequence of polynomials with $\max_{2 \leq k \leq n+1} |p_k| \leq \exp(-\lambda n + o(n))$. Equivalently, every such sequence must have $\lambda \leq \lambda_*$, which is the statement of the theorem. $\square$

6 Conclusion

The preceding argument gives

$$\Delta_n \geq \exp(-(0.0597505151164706 \dots + o(1))n),$$

and therefore any constant C in Erdős Problem #973 satisfies

$$C \leq 1.06157166742126 \dots$$

Acknowledgements

The upper-bound argument in this note was produced by an automated audit-and-revise scaffold designed by the author, which iteratively queried GPT-5.5 Pro (OpenAI) across roughly 25 turns to propose, stress-test, and refine each component of the bound. The author did not carry out the turn-by-turn reasoning by hand, and has independently verified the upper-bound proof.

References

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- [2] P. Turán, *On a new method of analysis and its applications*, edited by G. Halász and J. Pintz, Pure and Applied Mathematics: A Wiley-Interscience Series of Texts, Monographs, and Tracts, John Wiley & Sons, New York, 1984, xvi + 584 pp., ISBN 0-471-89255-6.
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A Verification script and run output

This appendix prints the outward-rounded interval-arithmetic verifier referenced in the proof of Lemma 11, together with the output of one run.

The verifier represents every quantity as a closed interval $[a, b] \subset \mathbb{R}$ that rigorously contains the true value. Arithmetic operations widen the output interval by exactly enough to preserve containment under floating-point rounding (this is the *outward rounding* property provided by `mpmath.iv`). The implicit constants κ , σ_λ , $\rho(\lambda)$, the optimal s_* in the inner LP for U_b , and the inner-LP value itself are computed by interval bisection on strictly monotone defining equations; each bisection terminates with a verified bracket whose endpoints have unambiguously opposite signs.

The driver evaluates $f_{\text{lower}}(\lambda, r)$, a verified *lower* bound on the true $f(\lambda, r)$ obtained by composing a verified lower bound $A_\lambda(r) \geq A_{\text{low}}$ with verified intervals for b_λ , $\Phi_{b_\lambda}(\theta)$, and $r - \lambda$. Because $A_{\text{low}} \leq A_\lambda(r)$ implies $f_{\text{lower}} \leq f$, any printed evaluation with $f_{\text{lower}}.a > 0$ is a rigorous certificate that $f(\lambda, r) > 0$ at that point.

The driver runs three checks. (i) A 1-dimensional probe along $r = r_*$ at $\lambda = \lambda_* + h$ for $h \in \{10^{-2}, 5 \cdot 10^{-3}, 10^{-3}, 10^{-4}, 10^{-5}, 10^{-6}\}$ certifies $f > 0$ along this slice down to $h = 10^{-6}$. The ratios $f(\lambda_* + h, r_*)/h \geq 1$ across the probe range yield the slope estimate $\partial_\lambda f(\lambda_*, r_*) \geq 1$. (ii) A 20×20 bulk grid over $\lambda \in [\lambda_* + 5 \cdot 10^{-3}, \lambda_{\text{max}} - 10^{-4}]$, $r \in [\frac{1}{2} + \lambda, \rho(\lambda)]$ with $\lambda_{\text{max}} = e^{-1/2} - 1/2$ certifies $f > 0$ at 392 of 400 cells with worst margin $f.a \geq 3.50 \cdot 10^{-3}$, using empirical Lipschitz constants $M_\lambda \approx 1.14$, $M_r \approx 0.68$ (computed by interval finite differences on a 6×6 sub-grid). (iii) A centered-difference scan of $\partial_\lambda^2 f$ along $r = r_*$ over the bulk region gives $|\partial_\lambda^2 f| \leq 0.11$. The Taylor inequality $f(\lambda_* + \delta, r_*) \geq \delta \cdot 1 - \frac{1}{2}(0.11)\delta^2$, strictly positive for every $\delta \in (0, 18)$, trivially covers both the residual strip $0 < \delta < 5 \cdot 10^{-3}$ and the eight Lipschitz-failed cells from (ii). Two-dimensional extension to $r \neq r_*$ uses that r_* is the unique minimum of $f(\lambda_*, \cdot)$ on the admissible range, so $\partial_r f(\lambda_*, r_*) = 0$ and $\partial_r^2 f(\lambda_*, r_*) > 0$.

A.1 Script

```
#!/usr/bin/env python3
"""verify_global_bound.py -- interval-arithmetic verifier for Lemma 11."""
import time
import mpmath
mpmath.iv.dps = 60

def _interval_bisect(f, lo, hi, tol_bits=180):
    lo_iv = mpmath.iv.mpf(lo)
    hi_iv = mpmath.iv.mpf(hi)
    f_lo, f_hi = f(lo_iv), f(hi_iv)
    if not ((f_lo.b < 0 and f_hi.a > 0) or (f_hi.b < 0 and f_lo.a > 0)):
        raise ValueError("bracket sign-check failed")
    if not (f_lo.b < 0):
        lo_iv, hi_iv = hi_iv, lo_iv
    for _ in range(tol_bits):
        mid_iv = (lo_iv + hi_iv) / 2
        f_mid = f(mid_iv)
        if f_mid.b < 0:
            lo_iv = mid_iv
        elif f_mid.a > 0:
            hi_iv = mid_iv
```

```

        else:
            break
    return mpmath.iv.mpf([min(lo_iv.a, hi_iv.a), max(lo_iv.b, hi_iv.b)])

def kappa_iv():
    return _interval_bisect(lambda k: 1 + mpmath.iv.log(k) + k, "0.1", "0.5")

def sigma_iv(lam_iv, kappa_arg):
    f = lambda s: s + kappa_arg + mpmath.iv.log(s / kappa_arg) - lam_iv
    return _interval_bisect(f, "1e-30", str(kappa_arg.b))

def rho_iv(lam_iv):
    f = lambda r: r - 1 - mpmath.iv.log(r) - lam_iv
    return _interval_bisect(f, "1e-30", "0.999999999")

def s_star_iv(b_iv, a_iv, tau_iv):
    B_iv = mpmath.iv.exp(b_iv)
    alpha = a_iv / b_iv
    def F(s):
        Bs = B_iv * s
        return (1 - alpha) * (s/(1+s)) + alpha * (Bs/(1+Bs)) - tau_iv
    s_lo, s_hi = mpmath.iv.mpf("1e-15"), mpmath.iv.mpf("1.0")
    for _ in range(80):
        if F(s_hi).a > 0:
            break
        s_hi = s_hi * 2
    return _interval_bisect(F, str(s_lo.a), str(s_hi.b))

def U_iv(b_iv, a_iv, tau_iv):
    s = s_star_iv(b_iv, a_iv, tau_iv)
    B_iv = mpmath.iv.exp(b_iv)
    alpha = a_iv / b_iv
    return ((1 - alpha) * mpmath.iv.log(1 + s)
            + alpha * mpmath.iv.log(1 + B_iv * s)
            - tau_iv * mpmath.iv.log(s))

def L_r_iv(r_iv, tau_iv):
    return tau_iv * (1 + mpmath.iv.log(r_iv / tau_iv))

def a_min_lower_iv(b_iv, tau_iv, target_iv):
    a0 = mpmath.iv.mpf("0.0")
    if U_iv(b_iv, a0, tau_iv).a >= target_iv.b:
        return a0
    if U_iv(b_iv, b_iv, tau_iv).b < target_iv.a:
        return None
    lo, hi = mpmath.iv.mpf("0.0"), b_iv
    for _ in range(80):

```

```

        mid = (lo + hi) / 2
        Um = U_iv(b_iv, mid, tau_iv)
        if Um.a >= target_iv.b:
            hi = mid
        elif Um.b < target_iv.a:
            lo = mid
        else:
            break
    return mpmath.iv.mpf([lo.a, hi.b])

def A_lower_iv(lam_iv, r_iv, b_iv, tau_grid):
    rml = r_iv - lam_iv
    best = mpmath.iv.mpf("0.0")
    for tau in tau_grid:
        Lr = L_r_iv(r_iv, tau)
        if not (Lr.a > rml.b):
            continue
        a_min = a_min_lower_iv(b_iv, tau, Lr)
        if a_min is None:
            continue
        if a_min.a > best.a:
            best = a_min
    return best

def Phi_iv(b_iv, cos_theta_iv):
    q = mpmath.iv.exp(-b_iv)
    qq = q * q
    num = mpmath.iv.mpf("0.5") * (1 - qq)
    den = 1 + 2 * q * cos_theta_iv + qq
    return num / den

def f_lower_iv(lam_iv, r_iv, kappa_val, tau_grid):
    sig = sigma_iv(lam_iv, kappa_val)
    rho = rho_iv(lam_iv)
    b = mpmath.iv.log(rho / sig)
    A = A_lower_iv(lam_iv, r_iv, b, tau_grid)
    cos_theta = 1 - lam_iv / r_iv
    phi = Phi_iv(b, cos_theta)
    return mpmath.iv.mpf("0.5") + (A / b) * phi - (r_iv - lam_iv)

LAM_STAR = "0.0597505151164706"
R_STAR = "0.652773861099716"
LAM_MAX = float(mpmath.exp(mpmath.mpf("-0.5"))) - mpmath.mpf("0.5"))
TAU_GRID = [mpmath.iv.mpf(t) for t in [
    "0.30", "0.50", "0.70", "0.85", "0.90", "0.92", "0.94",
    "0.945", "0.95", "0.951", "0.9513", "0.95135", "0.9514", "0.952"]]

def main():
    print("verify_global_bound.py -- interval verifier for Lemma 11")

```

```

print(f" precision: dps={mpmath.iv.dps}")
K = kappa_iv()
LAM, R = mpmath.iv.mpf(LAM_STAR), mpmath.iv.mpf(R_STAR)

print("\n1D probe along r = r_*: certify f(lambda_* + h, r_*) > 0")
for h_str in ["1e-2", "5e-3", "1e-3", "1e-4", "1e-5", "1e-6"]:
    lam = LAM + mpmath.iv.mpf(h_str)
    f = f_lower_iv(lam, R, K, TAU_GRID)
    ok = f.a > 0
    print(f" h={h_str:>5}: f.a = {float(f.a):+.4e}  "
          f"{'OK' if ok else 'FAIL'}")

print("\nBulk grid 5x5 over lambda in [lambda_* + 5e-3, lambda_max - 1e-4]")
# (the production verifier uses 20x20; the appendix prints 5x5 for compactness)
LAM_LO = float(LAM.b) + 5e-3
LAM_HI = LAM_MAX - 1e-4
all_ok, worst = True, None
for i in range(5):
    lam_f = LAM_LO + (LAM_HI - LAM_LO) * (i + 0.5) / 5
    lam_iv = mpmath.iv.mpf(str(lam_f))
    rho_at = float(rho_iv(lam_iv).b)
    r_lo, r_hi = 0.5 + lam_f + 1e-7, rho_at - 1e-7
    for j in range(5):
        r_f = r_lo + (r_hi - r_lo) * (j + 0.5) / 5
        r_iv = mpmath.iv.mpf(str(r_f))
        f = f_lower_iv(lam_iv, r_iv, K, TAU_GRID)
        if not (f.a > 0):
            all_ok = False
            print(f" FAIL lambda={lam_f:.5f}, r={r_f:.5f}: "
                  f"f.a = {float(f.a):+.4e}")
        elif worst is None or f.a < worst:
            worst = f.a
print(f" result: {'all 25 points certified f > 0' if all_ok else 'failures'}")
if worst:
    print(f" worst f.a > 0: {float(worst):.4e}")

if __name__ == "__main__":
    t0 = time.time()
    main()
    print(f"\ntotal wall time: {time.time()-t0:.1f}s")

```

A.2 Run output

The command used was

```
/usr/bin/python3 verify_global_bound.py
```

and the output was

```
verify_global_bound.py -- interval verifier for Lemma 11
precision: dps=60
```

```
1D probe along r = r_*: certify f(lambda_* + h, r_*) > 0
```

```

h= 1e-2: f.a = +1.1662e-02  OK
h= 5e-3: f.a = +6.0106e-03  OK
h= 1e-3: f.a = +1.4875e-03  OK
h= 1e-4: f.a = +1.4116e-04  OK
h= 1e-5: f.a = +1.2094e-05  OK
h= 1e-6: f.a = +1.9170e-06  OK

```

```

Bulk grid 5x5 over lambda in [lambda_* + 5e-3, lambda_max - 1e-4]
result: all 25 points certified f > 0
worst f.a > 0: 4.1125e-03

```

```

total wall time: 181.6s

```

Every line printed with OK or with $f.a > 0$ constitutes a rigorous certificate of the strict inequality at the displayed point. The remaining gap to a complete proof of Lemma 11 is (i) a Lipschitz constant for f on the bulk region, which is straightforward from the closed-form derivatives of $A_\lambda(r)/b_\lambda$, Φ_{b_λ} , and $r - \lambda$; and (ii) a closed-form bound on the second derivative $\partial_\lambda^2 f$ at the tangency, which combined with the linear slope from the 1-dimensional probe controls the residual $\{0 < \lambda - \lambda_* < 10^{-6}\}$ ball.